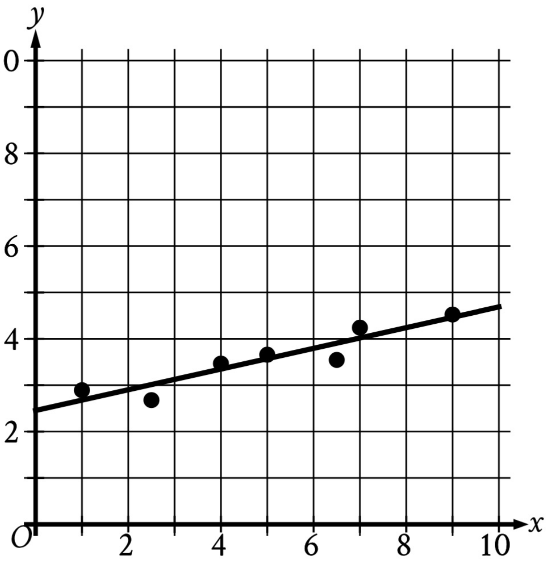
SAT 11

Math Module 3

Question #	ID			Answer
SAT11 M 3.1	a11ma301	Josie purchased a pass that cost $\$19.50$ to tour a nature preserve. She also purchased 10 identical stickers to give to her nieces and nephews. The pass and the stickers cost $\$62.00$ total. What was the cost of one sticker?	A. $\$1.95$ B. $\$4.25$ C. $\$6.20$ D. $\$8.15$	B
SAT11 M 3.2	a11ma302	$x+y=125$ $x+y+y=155$ The solution to the given system of equations is (x, y). What is the value of y?		30
SAT11 M 3.3	a11ma303	Leo goes to a packing store to buy containers and tape. Leo has $\$15$. Each container costs $\$1.87$ and each roll of tape costs $\$2.40$. Which inequality represents the relationship between the number of containers, c, and the number of rolls of tape, t, Leo can buy?	A. $1.87c+2.40t \leq 15$ B. $1.87c+2.40t \geq 15$ C. $2.40c+1.87t \leq 15$ D. $2.40c+1.87t \geq 15$	A

Question #	ID		Answer
SAT11 M 3.4	a11ma304		C

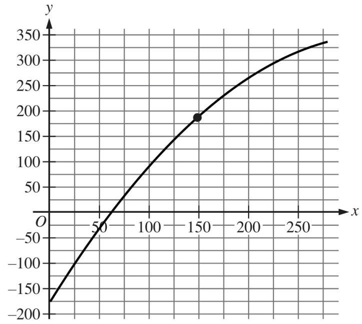
In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit?

A. -2.45
B. -0.22
C. 0.22
D. 2.45

Question #	ID	Answer
SAT11 M 3.5	a11ma305	B



The graph shows the estimated boiling point y, in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(149.02, 186.05)$?

A. A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.
B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.
C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.
D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

Question #	ID	Answer
SAT11 M 3.6	a11ma306	D

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

A. -4
B. 8
C. 42
D. 44

SAT 11

Math Module 3

Question #	ID		Answer
SAT11 M 3.7	a11ma307		100

Circle A in the xy-plane has the equation $(x+5)^2+(y-5)^2=25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy-plane is $(x+5)^2+(y-5)^2=k$, where k is a constant. What is the value of k?

SAT11 M 3.8 a11ma308 — Answer: **A**

The function f is defined by $f(x)=9(2x+3)$. For what value of x does $f(x)=63$?

A. 2
B. 5
C. 7
D. 30

SAT11 M 3.9 a11ma309 — Answer: **29**

$$x^2+7x+5=0$$

One solution to the given equation can be written as $x=\frac{-7+\sqrt{k}}{2}$, where k is a constant. What is the value of k?

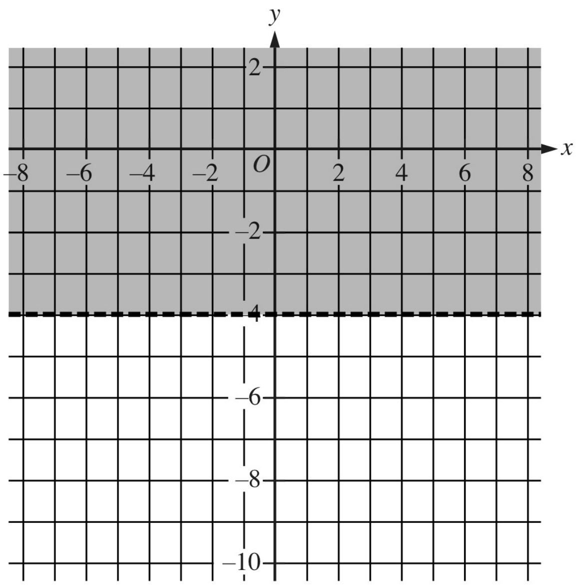
SAT11 M 3.10 a11ma310 — Answer: **A**

$$\left(9x^4+8x^3\right)-\left(7x^3+5x^2\right)-\left(3x^2+x\right)$$

The given expression can be written in the form $9x^4+x^3+ax^2-x$, where a is a constant. What is the value of a?

A. -8
B. -2
C. 2
D. 8

Question #	ID		Answer
SAT11 M 3.11	a11ma311		D



The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r?

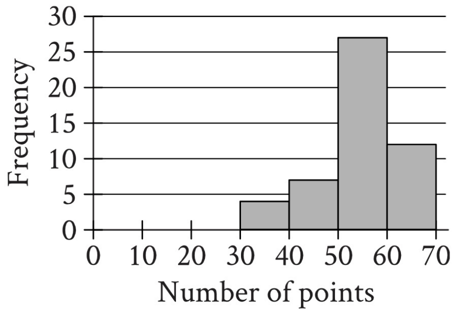
A. 15
B. 4
C. -4
D. -15

SAT11 M 3.12	a11ma312		A

Set K consists of all the positive integers that are less than or equal to 160, where none of the integers are repeated. If a number is selected at random from set K, what is the probability of selecting a number that is even and less than or equal to 50?

A. $\frac{5}{32}$
B. $\frac{3}{16}$
C. $\frac{5}{21}$
D. $\frac{5}{16}$

Question #	ID		Answer
SAT11 M 3.13	a11ma313		B



The histogram summarizes data set A, which represents the number of points per player earned by 50 players of a game. A new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. Which of the following must be true?

I. The median number of points per player for data set B is less than the median number of points per player for data set A.

II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

A. I only
B. II only
C. I and II
D. Neither I nor II

SAT11 M 3.14	a11ma314		C

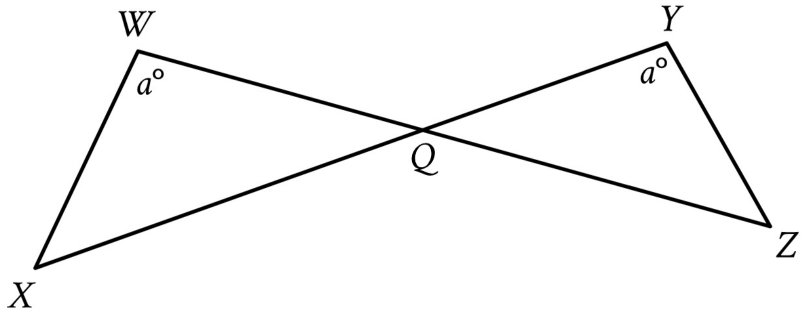
In a scale model of an office building, a length of 1 inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. If a length in the actual office building measures x feet, which expression represents the corresponding length, in inches, in the model in terms of x?

A. $\frac{3x}{(17)(12)}$
B. $\frac{17x}{(12)(3)}$
C. $\frac{3x}{17}$
D. $\frac{17x}{3}$

SAT11 M 3.15	a11ma315		41/81

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{3n-1}$, where n is a constant, what is the value of n?

Question #	ID			Answer
SAT11 M 3.16	a11ma316	$$g(x) = \frac{x^2 - x - a}{x^3 - x - b}$$ The function g is defined by the given equation, where a and b are constants. In the xy-plane, the graph of $y = g(x)$ passes through the point $(0, 22)$, and $g(-22) = 0$. What is the value of b?	A. 23 B. 22 C. -22 D. -23	A
SAT11 M 3.17	a11ma317	Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?	A. $(-18x + 280)^\circ$ B. $(-18x + 640)^\circ$ C. $(18x - 280)^\circ$ D. 180°	A
SAT11 M 3.18	a11ma318	 Note: Figure not drawn to scale. In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q, $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?	A. 18 B. 36 C. 120 D. 200	A

Question #	ID			Answer

SAT11 M 3.19 a11ma319 — Answer: C

$$2x + 9y = 7$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

I. $3x + 13.5y = 10.5$

II. $3x - 13.5y = 10.5$

A. I only

B. II only

C. I and II

D. Neither I nor II

SAT11 M 3.20 a11ma320 — Answer: B

A right rectangular prism has a base area of $24t$ square centimeters $\left(\mathrm{cm}^2\right)$. The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2, of the right rectangular prism?

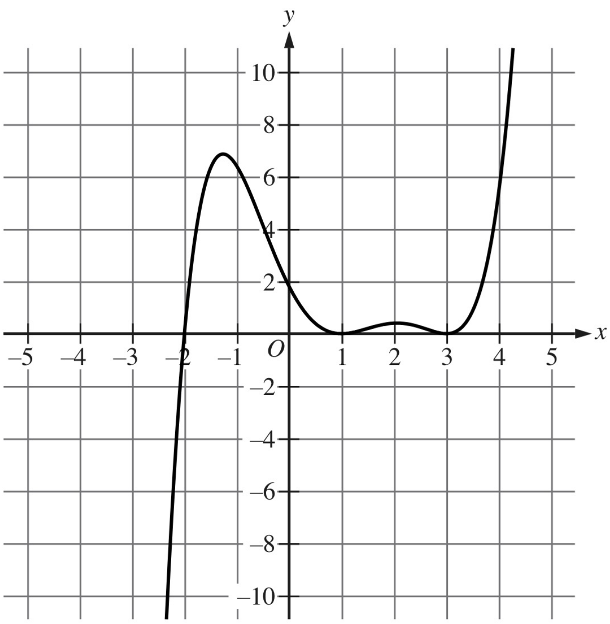
A. $48t + 160$

B. $318t + 80$

C. $1{,}968t + 80$

D. $360t$

SAT11 M 3.21 a11ma321 — Answer: D



The graph of $y = g(x)$ is shown. If $g(x + 28) = \frac{1}{10}(x - a)^2(x - b)^2(x - c)$, where a, b, and c are constants, which of the following could be the value of b?

A. 31

B. 3

C. -2

D. -25

Question #	ID	Answer
SAT11 M 3.22	a11ma322	60000

The weight of sample A is 468% of the weight of sample B, and the weight of sample A is 0.780% of the weight of sample C. If the weight of sample C is $p\%$ of the weight of sample B, what is the value of p?